Option A: Comprehensive Paper Plan for January 30th Deadline

You have **5 days** (Jan 25-30) to complete a combined paper covering both the physics simulator and HNN-GNN surrogate model.

Paper Structure & Outline

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│  PAPER TITLE (DRAFT):                                                                w       │
│  "A Physics-Informed Neural Network Framework for Industrial Heat Pump                       │
│   COP Prediction: Integrating Digital Twin Simulation with                                   │
│   Hamiltonian-Graph Neural Networks"                                                         │
│                                                                                              │
│  AUTHORS: [Your name], Airthium Team                                                         │
│  TARGET: arXiv/TechRxiv Preprint                                                             │
│  PAGE COUNT: ~12-15 pages                                                                    │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

Section-by-Section Outline

Abstract (~200 words)
```
Key points to cover:
• Problem: Real-time COP prediction for industrial heat pumps
• Solution: Two-part framework - physics simulator + neural surrogate
• Method: 17 integrated physics models + HNN-GNN architecture
• Results: COP distribution (mean 4.65, range 1.8-8.0), Carnot-compliant
• Impact: Faster inference while maintaining physical consistency
```

1. Introduction (~1.5 pages)
```
1.1 Background
   - Industrial heat pump importance for decarbonization
   - COP prediction challenges (nonlinear, multi-physics)
   - Gap: Need both accuracy (physics) AND speed (ML)
```

1.2 Related Work
- Traditional physics-based models (cite 3-5 papers)
- Pure ML approaches for HVAC (cite 3-5 papers)
- Physics-informed neural networks (cite 3-5 papers)
- HNN and GNN applications (cite 2-3 each)

1.3 Contributions
1. Comprehensive physics simulator integrating 17 models
2. Novel HNN-GNN hybrid architecture for temporal + topological learning
3. Physics-informed training with Carnot constraint
4. Validation showing realistic COP distribution
```

#### **2. Physics-Based Digital Twin** (~3 pages) - YOUR MAIN CONTRIBUTION
```
2.1 System Architecture
- 4-layer data structure diagram
- Flow from inputs → physics → derived → targets

2.2 Integrated Physics Models (17 models)
TABLE: Model name, purpose, key equations, source

Thermodynamics:
• RefrigerantProperties (CoolProp) - saturation curves, enthalpies
• CycleGenerator - vapor compression cycle states

Degradation:
• HeatExchangerFoulingModel - thermal resistance growth
• CompressorWearModel - efficiency degradation
• RefrigerantChargeModel - charge level effects
• FaultInjectionModel - fault scenarios

Environmental:
• AmbientConditionModel - weather effects, frost risk
• MultiZoneBuildingModel - load interactions

Control:
• ControlSystemModel - PID dynamics
• PartLoadCyclingModel - cycling losses

Dynamics:
• HamiltonianMechanicsModel - energy conservation
• ComponentInteractionModel - thermal/mass coupling

2.3 COP Calculation
   - Equation: COP = (Q_cond / W_comp) × degradation_factor
   - degradation_factor = Π(charge, fault, wear, ambient, part_load, cycling)
   - Carnot constraint enforcement

2.4 Validation
   - COP distribution statistics
   - Carnot efficiency check
   - Energy balance verification
```

#### **3. Neural Network Surrogate Model** (~2.5 pages)
```
3.1 Motivation for Neural Surrogate
   - Physics simulator accurate but slow (100-500ms)
   - Need <10ms for real-time monitoring
   - NN learns to approximate physics

3.2 HNN Component (Temporal Dynamics)
   - Input: 100 timesteps × 14 channels
   - Architecture: LSTM/GRU with Hamiltonian structure
   - Learns: startup transients, steady-state approach
   - Physics constraint: energy conservation

3.3 GNN Component (System Topology)
   - Input: 10 nodes × 35 features + 11 edges × 12 features
   - Architecture: Message-passing GNN
   - Learns: component interactions, heat/mass flow
   - Physics constraint: mass/energy balance at nodes

3.4 Hybrid Architecture
   - HNN encoder → temporal embedding
   - GNN encoder → topological embedding
   - Fusion layer → combined representation
   - Output head → COP prediction

3.5 Physics-Informed Loss
   - L = MSE(COP_pred, COP_true) + λ₁·Carnot_penalty + λ₂·Energy_penalty
```

#### **4. Experiments** (~2.5 pages)
```
4.1 Dataset Generation

- 10,000 cycles using physics simulator
- Stratified sampling across operating conditions
- Train/val/test split: 70/15/15

4.2 Training Setup
- Hardware: [your setup]
- Optimizer: Adam, lr=1e-4
- Epochs: 100
- Batch size: 32

4.3 Results
TABLE: Model performance comparison
- Physics simulator (baseline) - 100% "accuracy" by definition
- HNN only - MAE, R^2, inference time
- GNN only - MAE, R^2, inference time
- HNN-GNN hybrid - MAE, R^2, inference time

4.4 Analysis
- COP prediction error distribution
- Performance across different operating regimes
- Carnot limit compliance
- Inference latency comparison
```

#### **5. Discussion** (~1.5 pages)
```
5.1 When to Use Physics vs Neural Network
- Physics: Offline analysis, interpretability needed
- Neural: Real-time monitoring, edge deployment

5.2 Limitations
- Physics models based on synthetic data
- Need real Airthium data for transfer learning
- NN may extrapolate poorly outside training distribution

5.3 Future Work
- Real data integration
- Uncertainty quantification
- Online learning from operational data
```

#### **6. Conclusion** (~0.5 page)
```
- Summary of contributions

- Key results
- Broader impact for heat pump optimization
```

---

### What You Already Have (Ready to Use)

| Component | Status | File |
|---|---|---|
| Physics simulator | ✅ Complete | [`comprehensive_physics_generator_v2.py`](Synthetic%20Data%20Generation/comprehensive_physics_generator_v2.py) |
| 17 physics models | ✅ Loaded | [`src/`](Synthetic%20Data%20Generation/src/) directory |
| HNN-GNN model | ✅ Implemented | [`train_comprehensive_model.py`](Synthetic%20Data%20Generation/train_comprehensive_model.py) |
| Data loader | ✅ Ready | [`comprehensive_data_loader.py`](Synthetic%20Data%20Generation/comprehensive_data_loader.py) |
| 438 test cycles | ✅ Generated | [`industrial_dataset_v10/`](Synthetic%20Data%20Generation/industrial_dataset_v10/) |
| COP stats | ✅ Validated | mean=4.65, std=1.63, range [1.8, 8.0] |

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### What Needs To Be Done

| Task | Estimated Time | Priority |
|---|---|---|
| Generate 10,000 cycles | 1 hour (runs automatically) | HIGH |
| Train HNN-GNN model on 10K cycles | 2-4 hours | HIGH |
| Create architecture diagram (Figure 1) | 1-2 hours | HIGH |
| Create COP distribution plot (Figure 2) | 30 min | HIGH |
| Create model comparison table | 1 hour | HIGH |
| Write Introduction | 2-3 hours | MEDIUM |
| Write Physics Methods | 3-4 hours | HIGH |
| Write NN Methods | 2-3 hours | HIGH |
| Write Results | 2-3 hours | HIGH |
| Write Discussion/Conclusion | 1-2 hours | MEDIUM |
| Format and proofread | 2-3 hours | MEDIUM |

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```

Day-by-Day Timeline

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ DAY 1 (Jan 25 - Saturday): DATA & FIGURES                                                    │
├──────────────────────────────────────────────────────────────────────────────────────────────┤
│ Morning:                                                                                     │
│ • Generate 10,000 cycles (run script, let it execute)                                        │
│ • Start training HNN-GNN model                                                               │
│                                                                                              │
│ Afternoon:                                                                                   │
│ • Create Figure 1: System architecture diagram                                               │
│   - 4-layer structure                                                                        │
│   - Physics models integration                                                               │
│   - HNN-GNN architecture                                                                     │
│                                                                                              │
│ Evening:                                                                                     │
│ • Create Figure 2: COP distribution + Carnot efficiency                                      │
│ • Start paper outline in LaTeX/Word                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ DAY 2 (Jan 26 - Sunday): PHYSICS SECTION                                                     │
├──────────────────────────────────────────────────────────────────────────────────────────────┤
│ Morning:                                                                                     │
│ • Write Section 2.1: System Architecture                                                     │
│ • Write Section 2.2: Physics Models Table                                                    │
│                                                                                              │
│ Afternoon:                                                                                   │
│ • Write Section 2.3: COP Calculation methodology                                             │
│ • Write Section 2.4: Validation results                                                      │
│                                                                                              │
│ Evening:                                                                                     │
│ • Check NN training progress                                                                 │
│ • Create physics validation figures                                                          │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
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┌──┐
│ DAY 3 (Jan 27 - Monday): NEURAL NETWORK SECTION   │
├──┤
│ Morning: │
│ • Write Section 3.1-3.2: HNN component   │
│ • Write Section 3.3: GNN component   │
│   │
│ Afternoon: │
│ • Write Section 3.4: Hybrid architecture   │
│ • Write Section 3.5: Physics-informed loss   │
│ • Create NN architecture figure   │
│   │
│ Evening: │
│ • Collect training results   │
│ • Create performance comparison table   │
└──┘

┌──┐
│ DAY 4 (Jan 28 - Tuesday): RESULTS & ANALYSIS   │
├──┤
│ Morning: │
│ • Write Section 4: Experiments   │
│ • Create results tables and figures   │
│   │
│ Afternoon: │
│ • Write Introduction (now that you know results)   │
│ • Write Abstract   │
│   │
│ Evening: │
│ • Write Discussion section   │
│ • Write Conclusion   │
└──┘

┌──┐
│ DAY 5 (Jan 29 - Wednesday): POLISH   │
├──┤

```
│  Morning:                                                    │
│  • Complete any unfinished sections                          │
│  • Add references (aim for 25-35 citations)                  │
│                                                              │
│  Afternoon:                                                  │
│  • Review and revise entire paper                            │
│  • Check all figures and tables                              │
│                                                              │
│  Evening:                                                    │
│  • Final proofreading                                        │
│  • Prepare supplementary materials (code repo link)          │
└──────────────────────────────────────────────────────────────
──────────────────────┘

┌──────────────────────────────────────────────────────────────
──────────────────────┐
│  DAY 6 (Jan 30 - Thursday): SUBMISSION                       │
├──────────────────────────────────────────────────────────────
──────────────────────┤
│  Morning:                                                    │
│  • Final check                                               │
│  • Format for arXiv/TechRxiv                                 │
│                                                              │
│  Afternoon:                                                  │
│  • SUBMIT!                                                   │
└──────────────────────────────────────────────────────────────
──────────────────────┘
```

Key Figures Needed

1. **Figure 1: System Architecture** (Most important)
   - Shows 4-layer data structure
   - Physics model integration diagram
   - HNN-GNN architecture

2. **Figure 2: COP Distribution**
   - Histogram of generated COP values
   - Carnot efficiency scatter plot
   - Show it's below Carnot limit

3. **Figure 3: Physics Model Coverage**

   - Table or diagram of 17 models
   - What each model computes

4. **Figure 4: Neural Network Architecture**
   - HNN branch (temporal)
   - GNN branch (topological)
   - Fusion and output

5. **Figure 5: Results**
   - Prediction vs actual COP scatter
   - Error distribution
   - Inference time comparison

Immediate Next Steps

1. **Today (Jan 25)**: Start generating 10,000 cycles:
   ```bash
   cd "Synthetic Data Generation"
   python comprehensive_physics_generator_v2.py --n_cycles 10000 --output_dir paper_dataset
   ```

2. **Train the model** on generated data

3. **Create the architecture diagram** - this is the centerpiece of your paper

Would you like me to:
1. Help create the detailed architecture diagram in Mermaid/ASCII for Figure 1?
2. Provide a LaTeX template for the paper?
3. Draft specific sections of the paper?
4. Help prepare the code for the 10,000 cycle generation?

┌──┐
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4. Experiments (~2.5 pages)
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- Uncertainty quantification
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6. Conclusion (~0.5 page)
- Summary of contributions
- Key results
- Broader impact for heat pump optimization

Key Figures Needed
Figure 1: System Architecture (Most important)

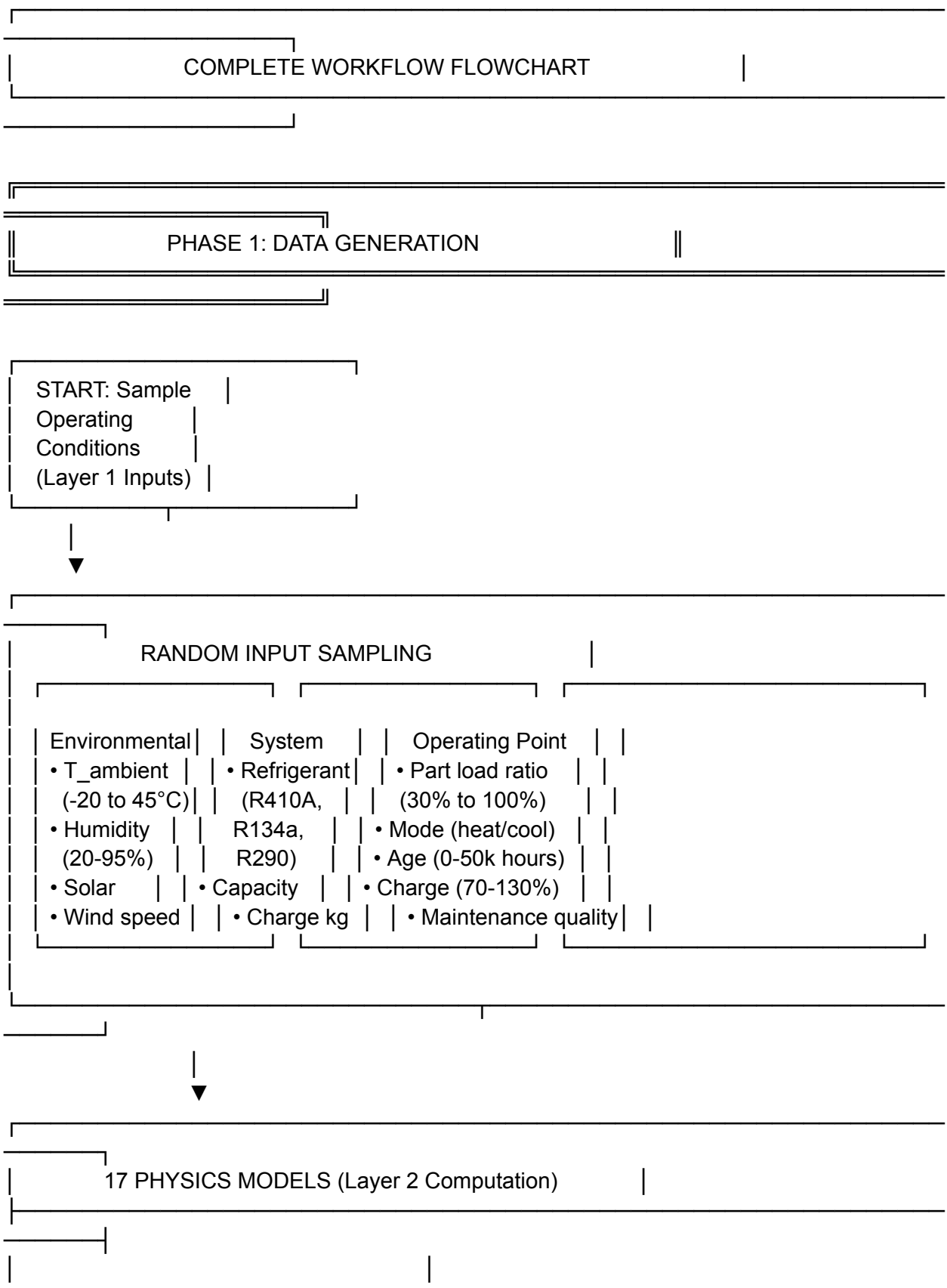
Shows 4-layer data structure
Physics model integration diagram
HNN-GNN architecture
Figure 2: COP Distribution

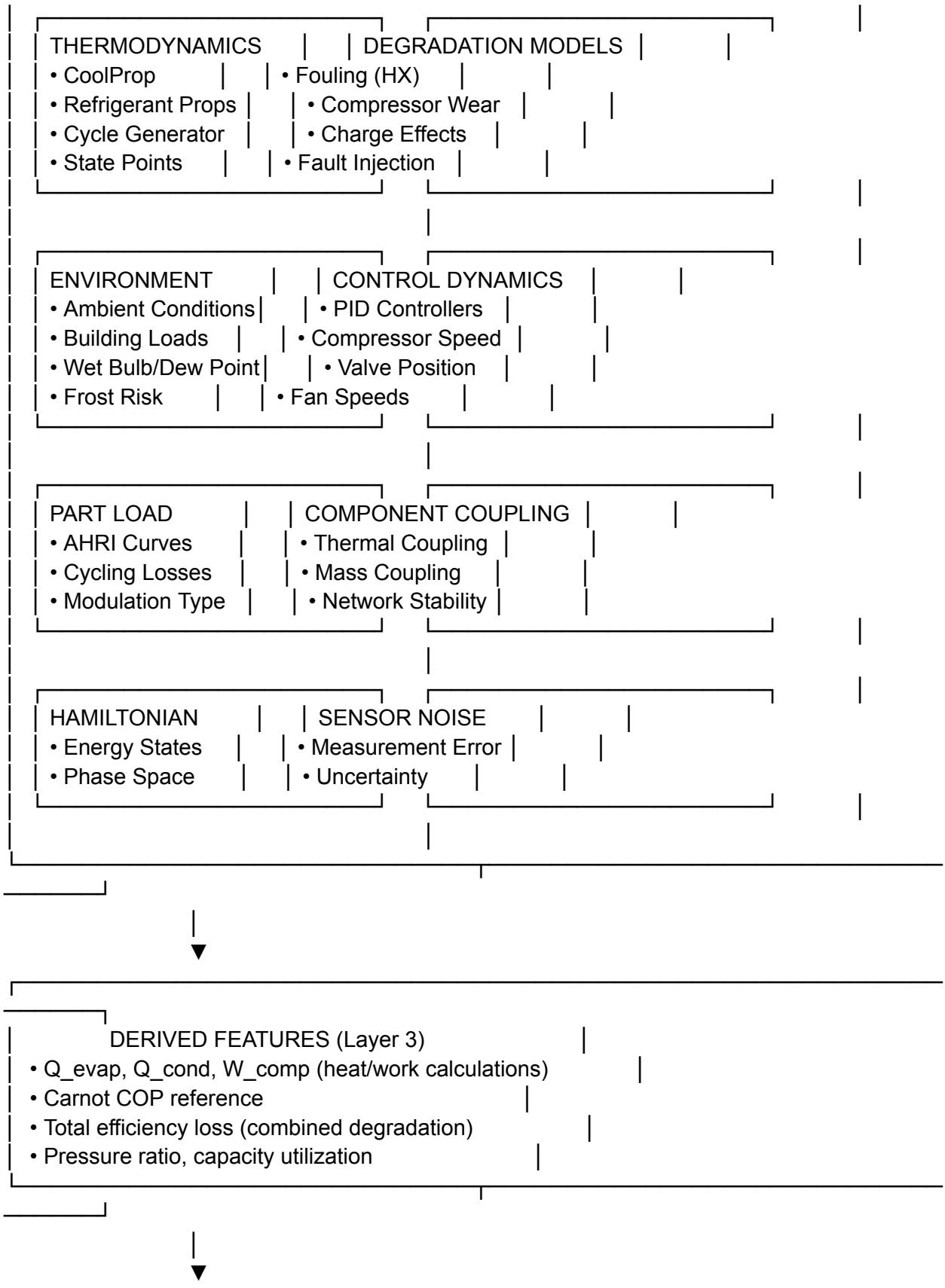
Histogram of generated COP values
Carnot efficiency scatter plot
Show it's below Carnot limit
Figure 3: Physics Model Coverage

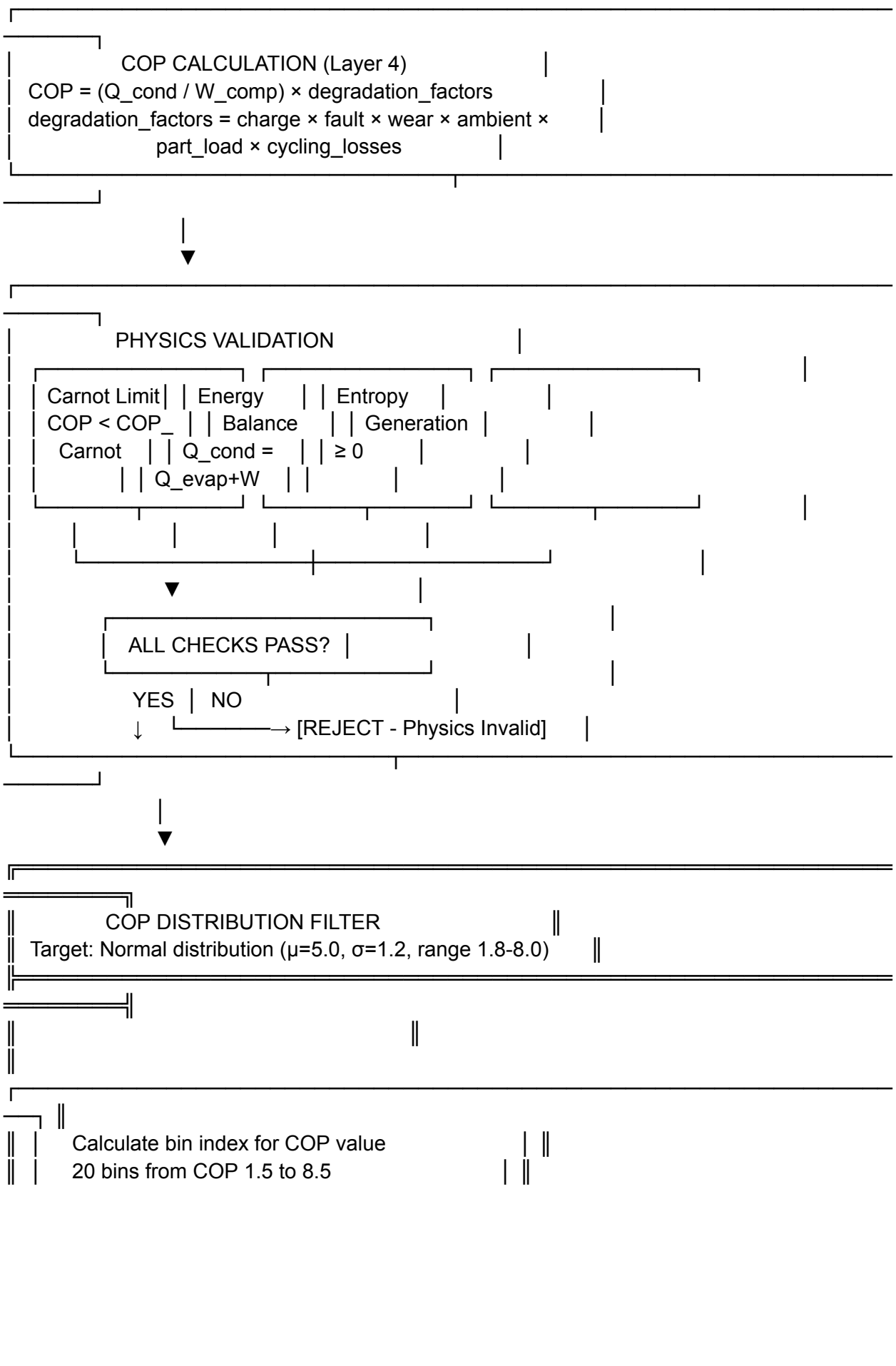
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What each model computes
Figure 4: Neural Network Architecture

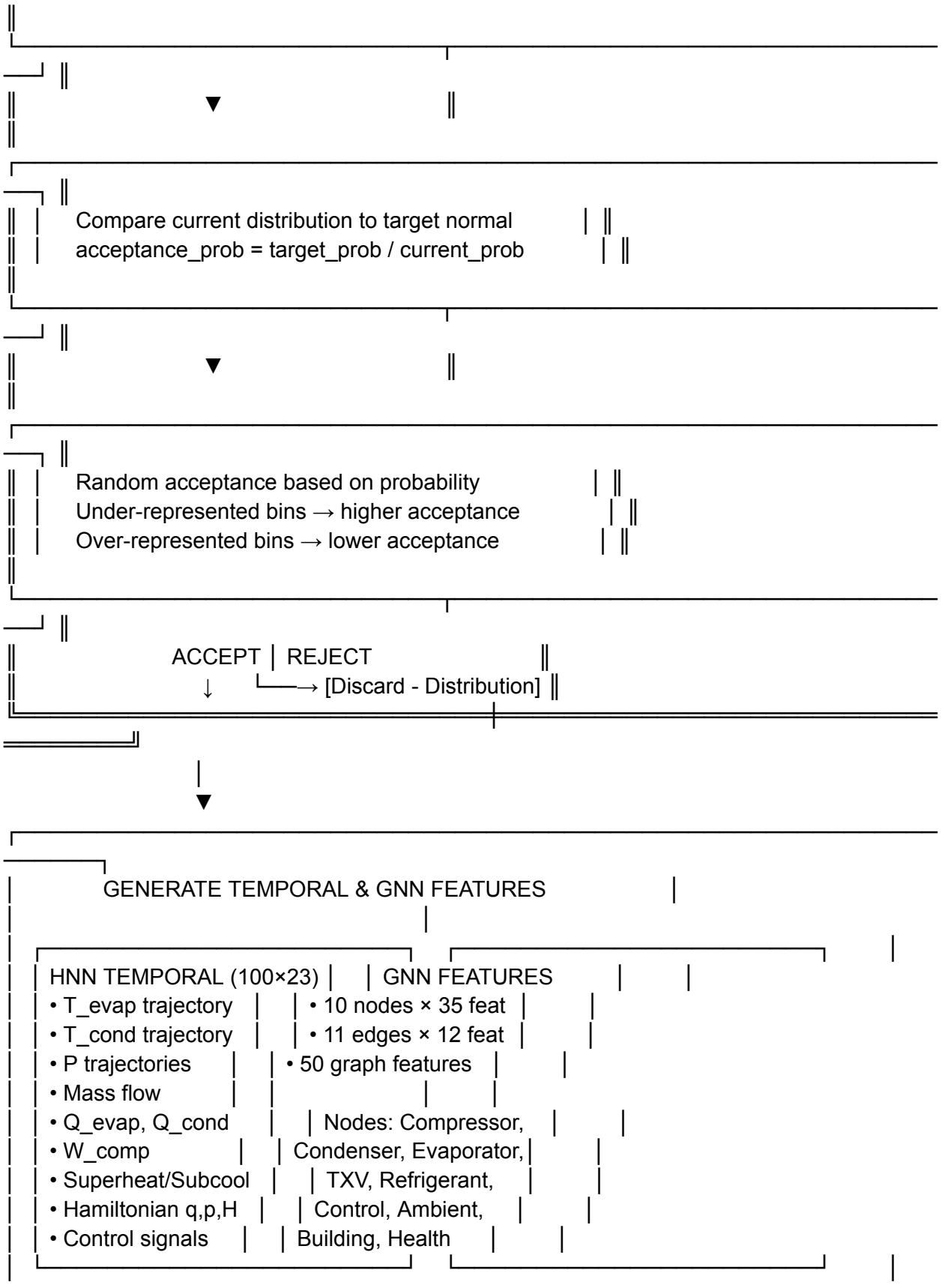
HNN branch (temporal)
GNN branch (topological)
Fusion and output
Figure 5: Results

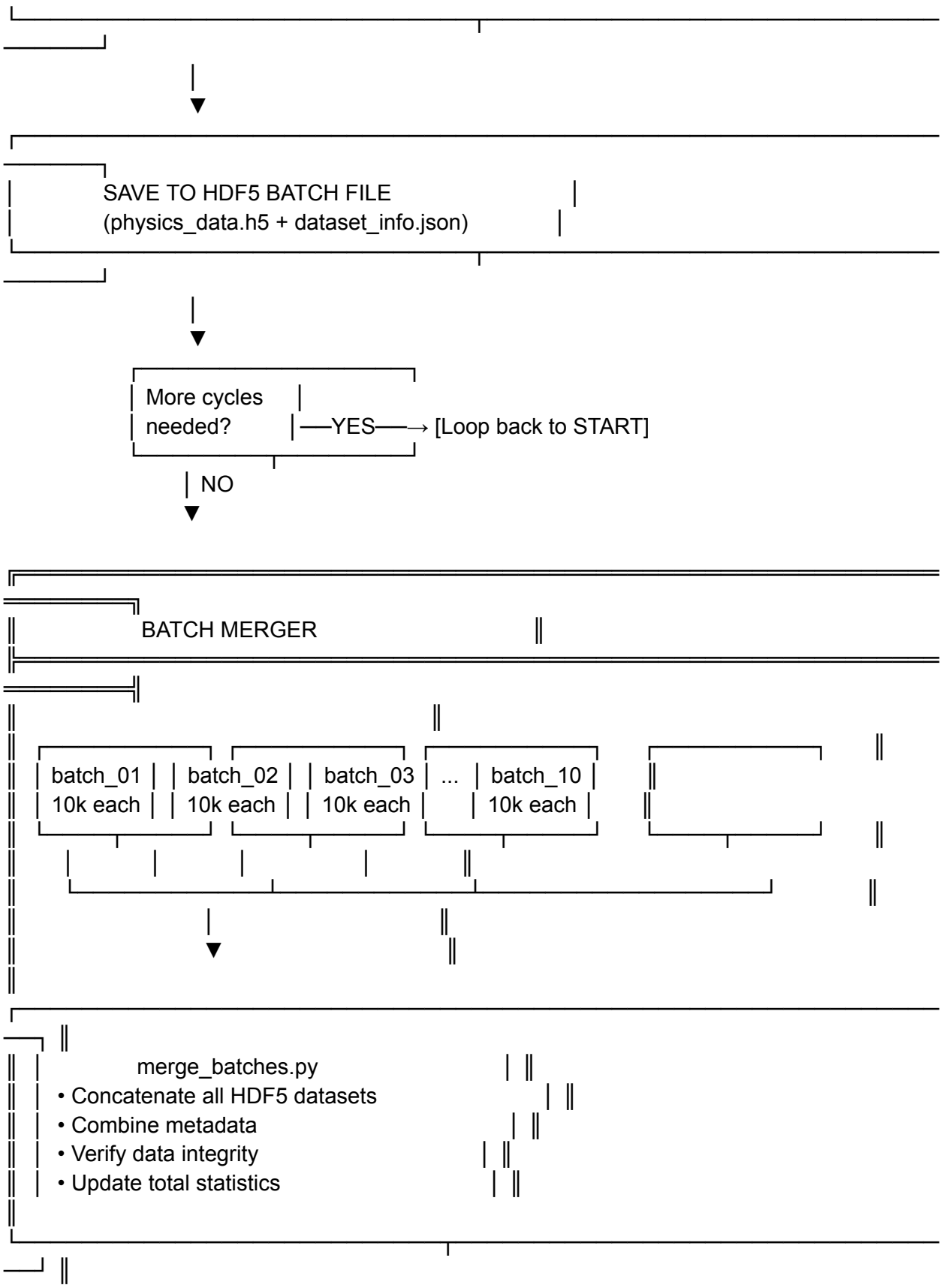
Prediction vs actual COP scatter
Error distribution
Inference time comparison

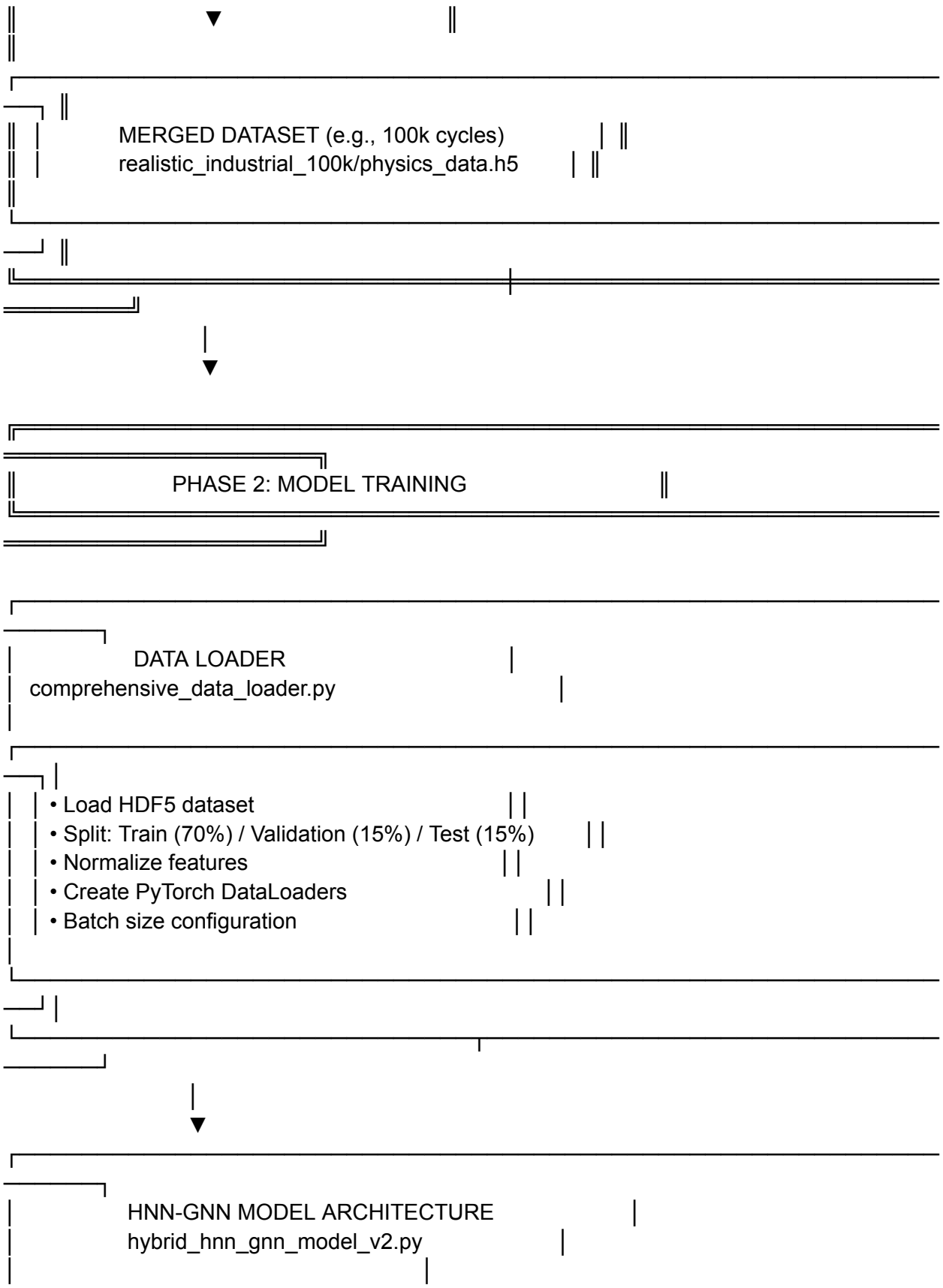

COMPLETE WORKFLOW FLOWCHART
PHASE 1: DATA GENERATION
START: Sample Operating Conditions (Layer 1 Inputs)
RANDOM INPUT SAMPLING
Environmental
• T_ambient (-20 to 45°C)
• Humidity (20-95%)
• Solar
• Wind speed
System
• Refrigerant (R410A, R134a, R290)
• Capacity
• Charge kg
Operating Point
• Part load ratio (30% to 100%)
• Mode (heat/cool)
• Age (0-50k hours)
• Charge (70-130%)
• Maintenance quality
17 PHYSICS MODELS (Layer 2 Computation)

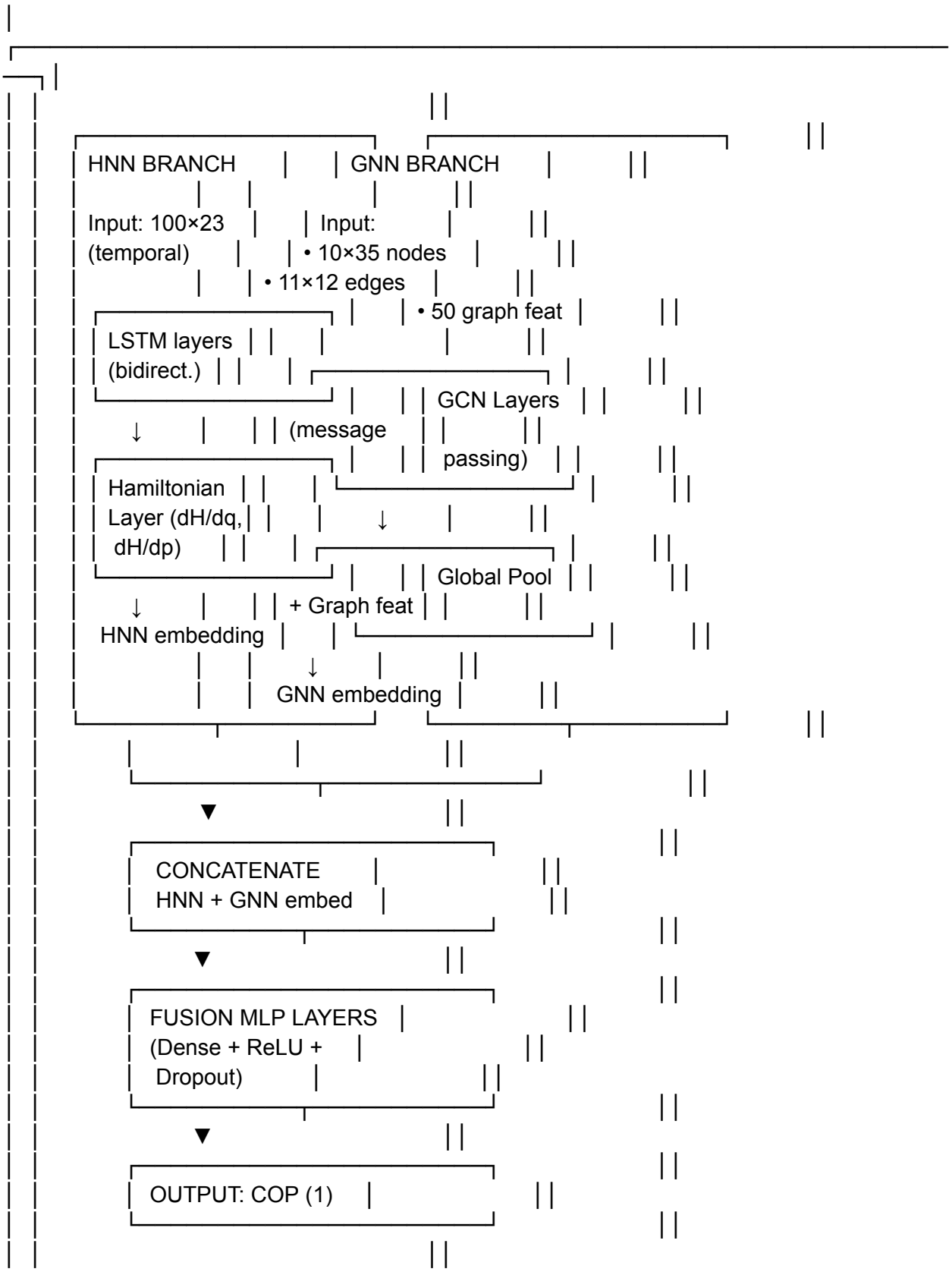

THERMODYNAMICS
• CoolProp
• Refrigerant Props
• Cycle Generator
• State Points
DEGRADATION MODELS
• Fouling (HX)
• Compressor Wear
• Charge Effects
• Fault Injection
ENVIRONMENT
• Ambient Conditions
• Building Loads
• Wet Bulb/Dew Point
• Frost Risk
CONTROL DYNAMICS
• PID Controllers
• Compressor Speed
• Valve Position
• Fan Speeds
PART LOAD
• AHRI Curves
• Cycling Losses
• Modulation Type
COMPONENT COUPLING
• Thermal Coupling
• Mass Coupling
• Network Stability
HAMILTONIAN
• Energy States
• Phase Space
SENSOR NOISE
• Measurement Error
• Uncertainty
DERIVED FEATURES (Layer 3)
• Q_evap, Q_cond, W_comp (heat/work calculations)
• Carnot COP reference
• Total efficiency loss (combined degradation)
• Pressure ratio, capacity utilization

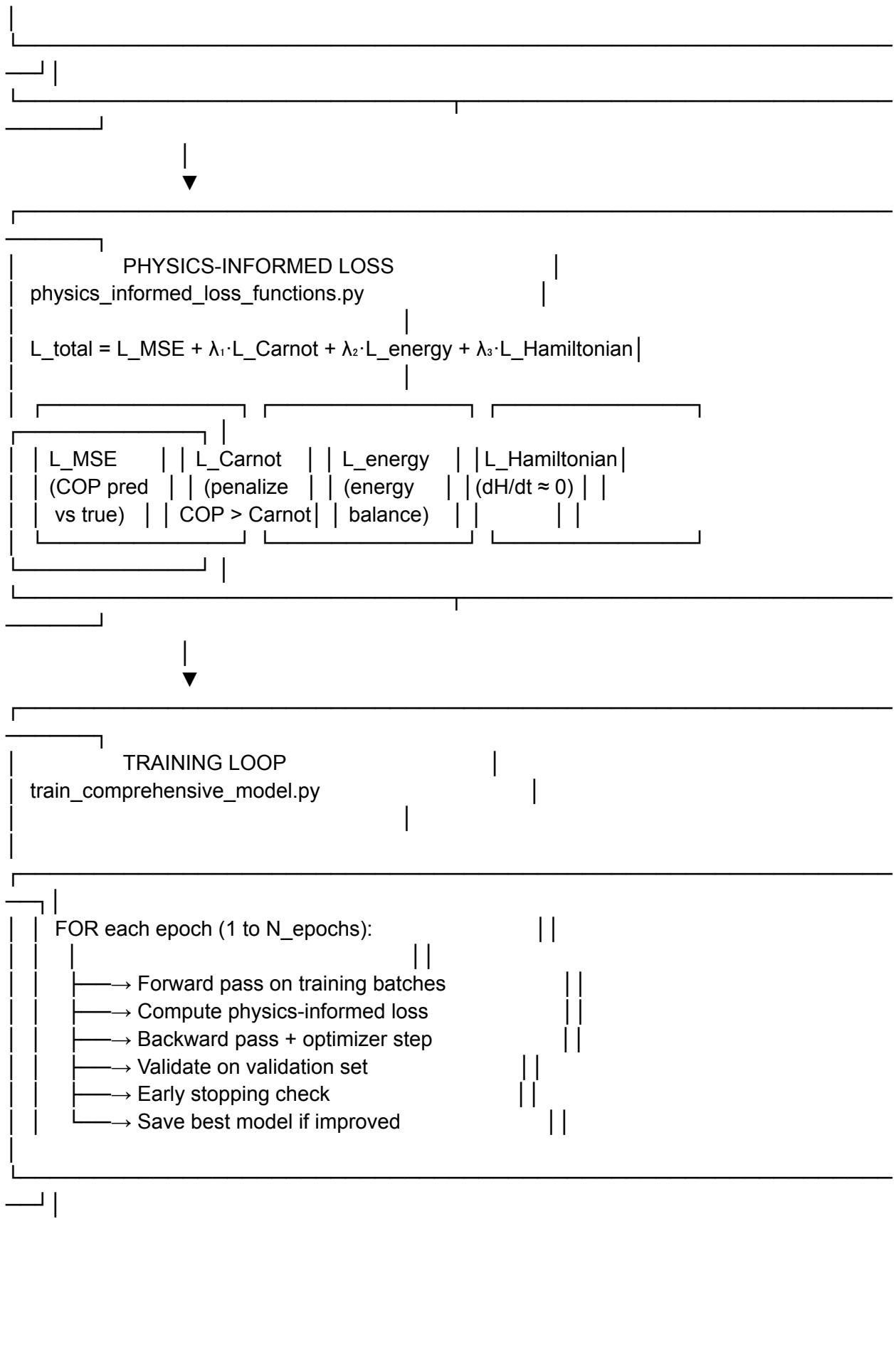

COP CALCULATION (Layer 4)
COP = (Q_cond / W_comp) × degradation_factors
degradation_factors = charge × fault × wear × ambient × part_load × cycling_losses
PHYSICS VALIDATION
Carnot Limit
COP < COP_Carnot
Energy Balance
Q_cond = Q_evap+W
Entropy Generation
≥ 0
ALL CHECKS PASS?
YES
NO
[REJECT - Physics Invalid]
COP DISTRIBUTION FILTER
Target: Normal distribution (μ=5.0, σ=1.2, range 1.8-8.0)
Calculate bin index for COP value
20 bins from COP 1.5 to 8.5

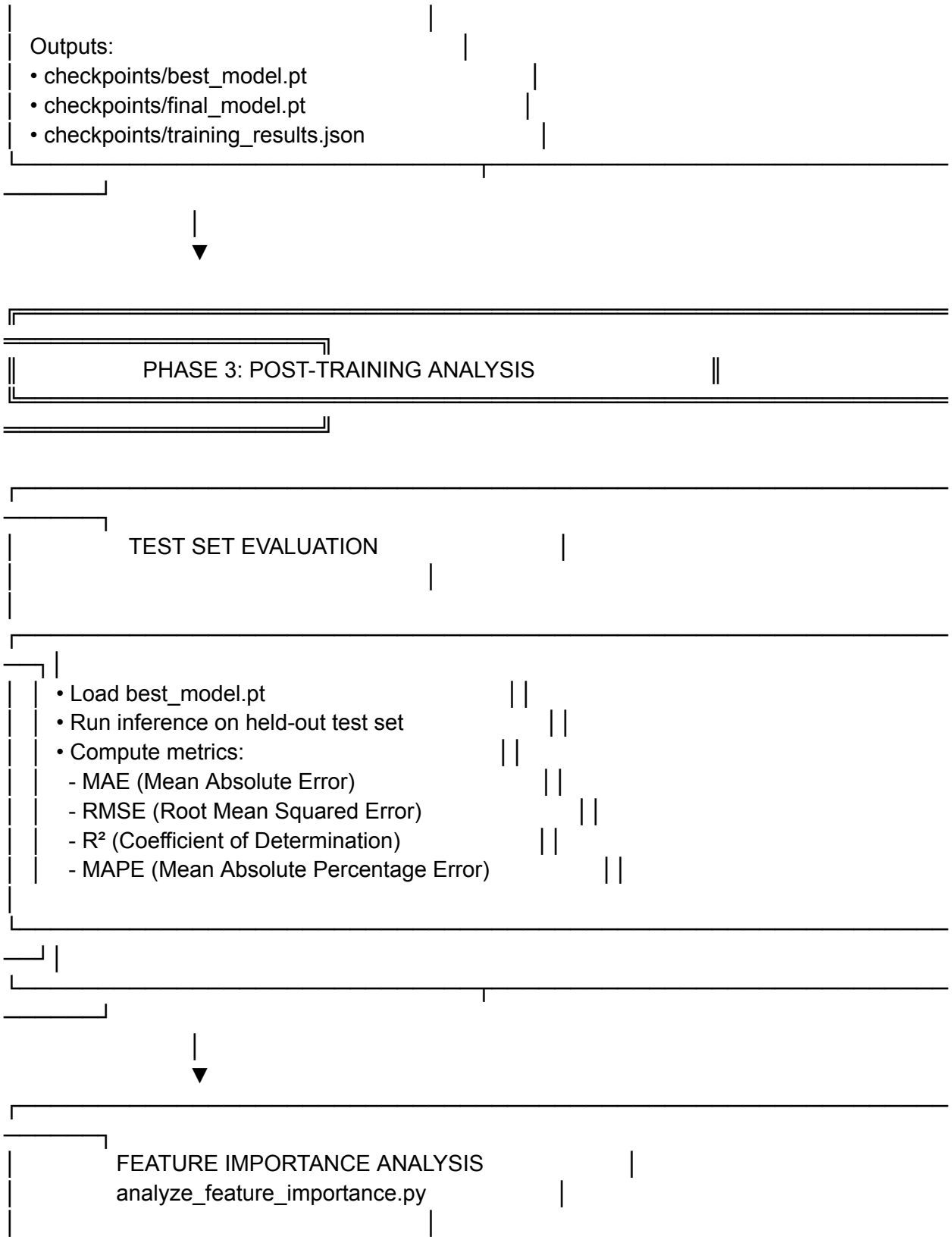

Compare current distribution to target normal
acceptance_prob = target_prob / current_prob
Random acceptance based on probability
Under-represented bins → higher acceptance
Over-represented bins → lower acceptance
ACCEPT
REJECT
[Discard - Distribution]
GENERATE TEMPORAL & GNN FEATURES
HNN TEMPORAL (100×23)
• T_evap trajectory
• T_cond trajectory
• P trajectories
• Mass flow
• Q_evap, Q_cond
• W_comp
• Superheat/Subcool
• Hamiltonian q,p,H
• Control signals
GNN FEATURES
• 10 nodes × 35 feat
• 11 edges × 12 feat
• 50 graph features
Nodes: Compressor,
Condenser, Evaporator,
TXV, Refrigerant,
Control, Ambient,
Building, Health

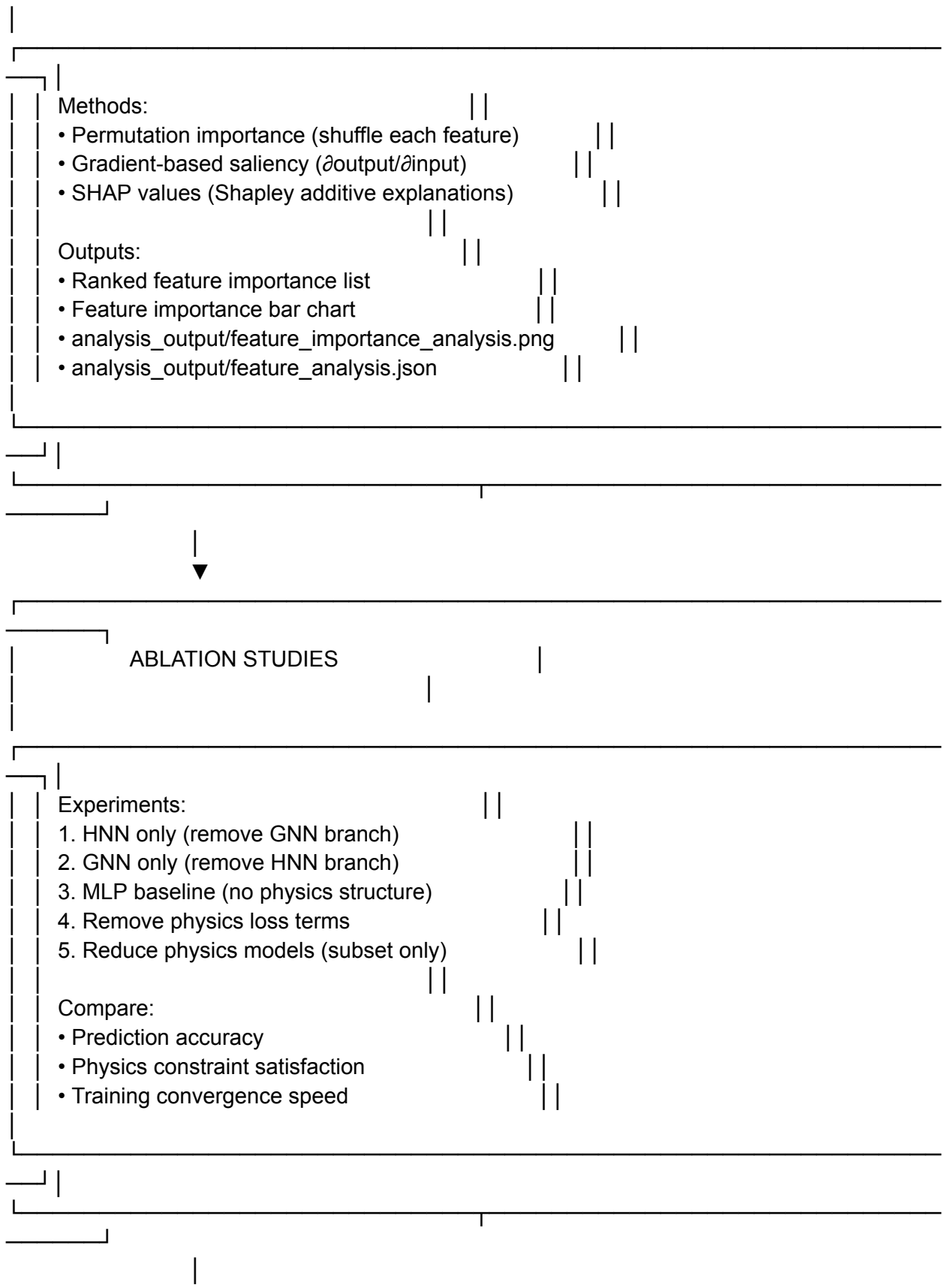

SAVE TO HDF5 BATCH FILE
(physics_data.h5 + dataset_info.json)
More cycles needed?
YES
[Loop back to START]
NO
BATCH MERGER
batch_01
10k each
batch_02
10k each
batch_03
10k each
...
batch_10
10k each
merge_batches.py
• Concatenate all HDF5 datasets
• Combine metadata
• Verify data integrity
• Update total statistics

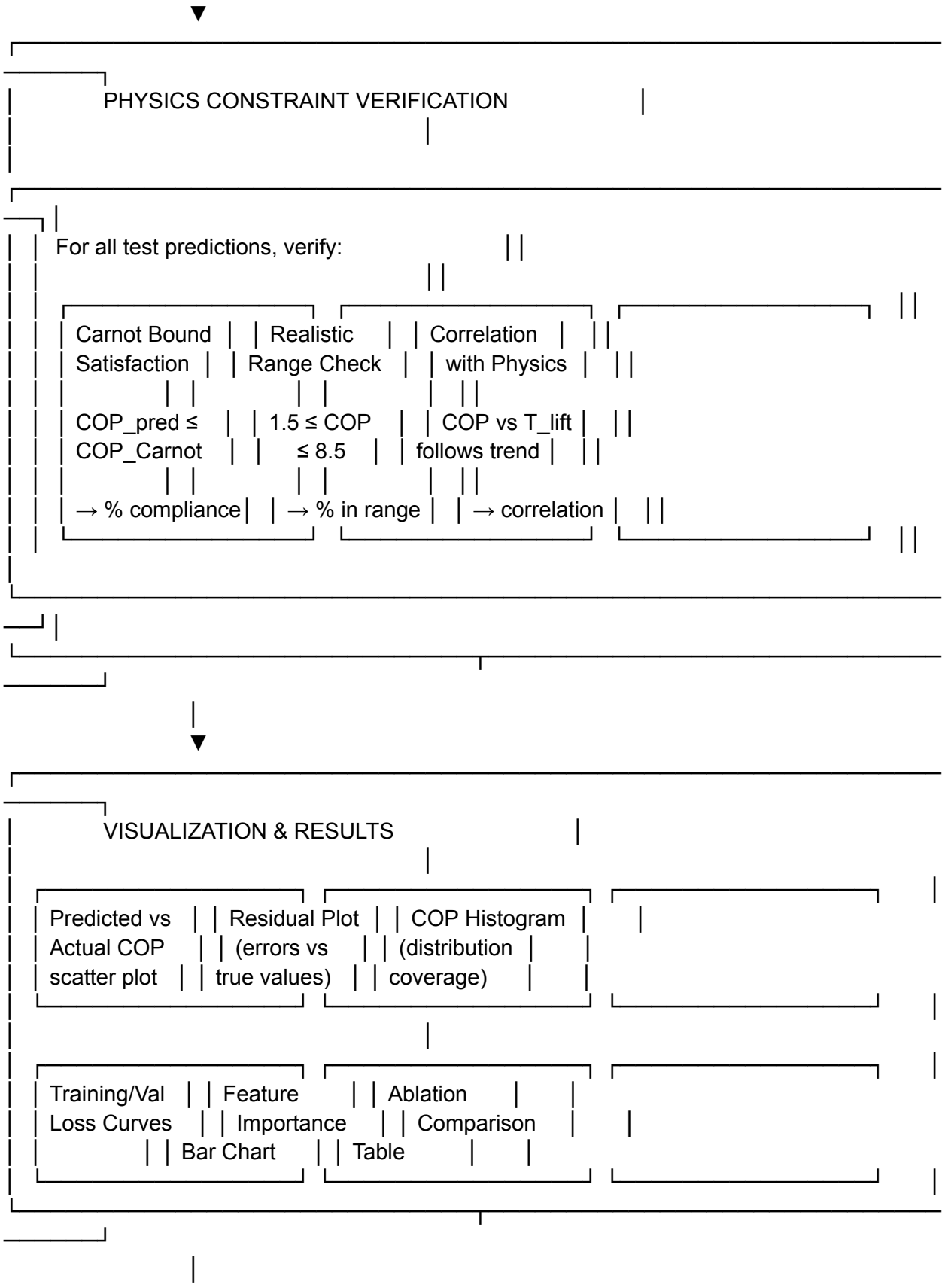

MERGED DATASET (e.g., 100k cycles)
realistic_industrial_100k/physics_data.h5
PHASE 2: MODEL TRAINING
DATA LOADER
comprehensive_data_loader.py
• Load HDF5 dataset
• Split: Train (70%) / Validation (15%) / Test (15%)
• Normalize features
• Create PyTorch DataLoaders
• Batch size configuration
HNN-GNN MODEL ARCHITECTURE
hybrid_hnn_gnn_model_v2.py

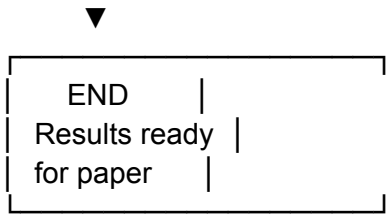

HNN BRANCH
Input: 100×23
(temporal)
LSTM layers
(bidirect.)
Hamiltonian
Layer (dH/dq,
dH/dp)
HNN embedding
GNN BRANCH
Input:
• 10×35 nodes
• 11×12 edges
• 50 graph feat
GCN Layers
(message
passing)
Global Pool
+ Graph feat
GNN embedding
CONCATENATE
HNN + GNN embed
FUSION MLP LAYERS
(Dense + ReLU +
Dropout)
OUTPUT: COP (1)


PHYSICS-INFORMED LOSS
physics_informed_loss_functions.py
L_total = L_MSE + λ₁·L_Carnot + λ₂·L_energy + λ₃·L_Hamiltonian
L_MSE
(COP pred vs true)
L_Carnot
(penalize COP > Carnot
L_energy
(energy balance)
L_Hamiltonian
(dH/dt ≈ 0)
TRAINING LOOP
train_comprehensive_model.py
FOR each epoch (1 to N_epochs):
Forward pass on training batches
Compute physics-informed loss
Backward pass + optimizer step
Validate on validation set
Early stopping check
Save best model if improved


Outputs:
• checkpoints/best_model.pt
• checkpoints/final_model.pt
• checkpoints/training_results.json
PHASE 3: POST-TRAINING ANALYSIS
TEST SET EVALUATION
• Load best_model.pt
• Run inference on held-out test set
• Compute metrics:
- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R² (Coefficient of Determination)
- MAPE (Mean Absolute Percentage Error)
FEATURE IMPORTANCE ANALYSIS
analyze_feature_importance.py


Methods:
• Permutation importance (shuffle each feature)
• Gradient-based saliency (∂output/∂input)
• SHAP values (Shapley additive explanations)
Outputs:
• Ranked feature importance list
• Feature importance bar chart
• analysis_output/feature_importance_analysis.png
• analysis_output/feature_analysis.json
ABLATION STUDIES
Experiments:
1. HNN only (remove GNN branch)
2. GNN only (remove HNN branch)
3. MLP baseline (no physics structure)
4. Remove physics loss terms
5. Reduce physics models (subset only)
Compare:
• Prediction accuracy
• Physics constraint satisfaction
• Training convergence speed


PHYSICS CONSTRAINT VERIFICATION
For all test predictions, verify:
Carnot Bound Satisfaction
COP_pred ≤ COP_Carnot
→ % compliance
Realistic Range Check
1.5 ≤ COP ≤ 8.5
→ % in range
Correlation with Physics
COP vs T_lift follows trend
→ correlation
VISUALIZATION & RESULTS
Predicted vs Actual COP scatter plot
Residual Plot (errors vs true values)
COP Histogram (distribution coverage)
Training/Val Loss Curves
Feature Importance Bar Chart
Ablation Comparison Table



Key Components Summary:

Phase 1: Data Generation

1. Random Input Sampling - Layer 1 operating conditions
2. 17 Physics Models - Layer 2 computations
3. Derived Features - Layer 3 calculations
4. COP Calculation - Layer 4 with degradation
5. Physics Validation - Carnot, energy, entropy checks
6. Distribution Filter - Shapes output to target normal distribution
7. Temporal/GNN Feature Generation - 2,832 features total
8. Batch Generation - Multiple batches (10k each)
9. Batch Merger - Combines into final dataset

Phase 2: Model Training

1. Data Loader - Train/Val/Test split, normalization
2. HNN-GNN Architecture - Dual branch with fusion
3. Physics-Informed Loss - Multiple constraint terms
4. Training Loop - Early stopping, checkpointing

Phase 3: Post-Training Analysis

1. Test Evaluation - MAE, RMSE, R^2
2. Feature Importance - Permutation/gradient/SHAP
3. Ablation Studies - Component contribution
4. Physics Verification - Constraint satisfaction
5. Visualizations - Publication-ready figures

Here's the complete workflow flowchart for your paper methodology:

PHASE 1: DATA GENERATION

```
START: Random Input Sampling (Layer 1)
├── Environmental: T_ambient (-20 to 45°C), humidity (20-95%), solar, wind
├── System: Refrigerant (R410A/R134a/R290), capacity, charge
└── Operating: Part load (30-100%), mode, age (0-50k hrs), maintenance
        │
        ▼
17 PHYSICS MODELS (Layer 2)
├── Thermodynamics: CoolProp, Refrigerant Properties, Cycle Generator
├── Degradation: Fouling, Compressor Wear, Charge Effects, Faults
├── Environment: Ambient Conditions, Building Loads, Frost Risk
├── Control: PID Controllers, Compressor Speed, Valve Position, Fan Speeds
├── Part Load: AHRI Curves, Cycling Losses, Modulation Types
└── Coupling: Thermal/Mass Coupling, Network Stability, Hamiltonian
        │
        ▼
DERIVED FEATURES (Layer 3)
├── Q_evap, Q_cond, W_comp calculations
├── Carnot COP reference
└── Total efficiency loss (combined degradation)
        │
        ▼
COP CALCULATION (Layer 4)
COP = (Q_cond/W_comp) × charge × fault × wear × ambient × part_load × cycling
        │
        ▼
PHYSICS VALIDATION
├── Carnot Limit: COP < COP_Carnot
├── Energy Balance: Q_cond = Q_evap + W_comp
├── Entropy Generation ≥ 0
└── PASS? ──NO──→ [REJECT - Physics Invalid]
        │ YES
        ▼
COP DISTRIBUTION FILTER
├── Target: Normal distribution (μ=5.0, σ=1.2)
├── 20 bins from COP 1.5 to 8.5
├── acceptance_prob = target_prob / current_prob
└── ACCEPT? ──NO──→ [Discard - Distribution]
        │ YES
        ▼
GENERATE FEATURES
├── HNN: 100 timesteps × 23 channels (temporal trajectories)
```

```
└── GNN: 10 nodes × 35 feat + 11 edges × 12 feat + 50 graph feat
        │
        ▼
SAVE BATCH (HDF5) ──→ [Loop if more cycles needed]
        │
        ▼
BATCH MERGER
batch_01 + batch_02 + ... + batch_10 → merged_dataset (100k cycles)
```

PHASE 2: MODEL TRAINING

```
DATA LOADER
├── Load HDF5, Split 70/15/15 (Train/Val/Test)
├── Normalize features
└── Create PyTorch DataLoaders
        │
        ▼
HNN-GNN ARCHITECTURE
├── HNN Branch: Input (100×23) → LSTM → Hamiltonian Layer → Embedding
├── GNN Branch: Input (10×35 nodes, 11×12 edges, 50 graph) → GCN → Pool → Embedding
├── Fusion: Concatenate → MLP Layers → Output (COP)
        │
        ▼
PHYSICS-INFORMED LOSS
L_total = L_MSE + λ₁·L_Carnot + λ₂·L_energy + λ₃·L_Hamiltonian
        │
        ▼
TRAINING LOOP
├── Forward pass, Loss computation, Backward pass
├── Validation check, Early stopping
└── Save: best_model.pt, final_model.pt, training_results.json
```

PHASE 3: POST-TRAINING ANALYSIS

```
TEST EVALUATION
├── MAE, RMSE, R², MAPE
        │
        ▼
FEATURE IMPORTANCE
├── Permutation importance
```

```
├── Gradient-based saliency
├── SHAP values
└── Output: feature_importance_analysis.png
        │
        ▼
ABLATION STUDIES
├── HNN only, GNN only, MLP baseline
├── Remove physics loss terms
└── Compare accuracy and convergence
        │
        ▼
PHYSICS VERIFICATION
├── Carnot bound satisfaction %
├── Realistic range check (1.5-8.5)
└── COP vs T_lift correlation
        │
        ▼
VISUALIZATIONS
├── Predicted vs Actual scatter
├── Residual plot
├── COP histogram
├── Training curves
└── Feature importance chart
```

You can recreate this in any diagramming tool (draw.io, Lucidchart, PowerPoint) with boxes and arrows following this structure.